

Hamza Rafi

Electrical & Computer Engineering · Rutgers University

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EDUCATION

Rutgers University, New Brunswick, NJ

2024-Present

Ph.D. in Electrical and Computer Engineering

Specialization: Signal and Information Processing

Advisors: [Prof. Waheed U. Bajwa](#) and [Prof. Mert Gurbuzbalaban](#)

Lahore University of Management Sciences, Lahore, Pakistan

2019-2023

B.S. in Electrical Engineering

Graduated with Distinction

Advisor: [Prof. Muhammad Tahir](#)

Senior-Year Project: Forecasting of Taxi Demand for an Urban Centre

RESEARCH INTERESTS

First-order optimization methods; decision-making under uncertainty; robust and efficient AI systems.

RESEARCH EXPERIENCE

Robust Optimization for First-Order Methods under Delay

2026-Present

Rutgers University

- Analyzing the robustness of momentum-based first-order methods under stale-gradient and delay models.
- Extending existing analyses from gradient descent to generalized momentum methods in quadratic optimization settings.
- Deriving theoretical guarantees and experimentally verifying results through numerical simulations.
- Developed a reproducible implementation pipeline, with code available on GitHub.

Efficient Decision-Making under Uncertainty

2025-Present

Rutgers University

- Exploring statistically grounded approaches for learning-to-defer and uncertainty-aware prediction.
- Studying decision rules that balance model confidence, expert deferral, and computational efficiency.
- Developing experimental pipelines for evaluating uncertainty-guided classification and deferral mechanisms.

Training-Free Textual Guidance for Improving Stable Diffusion 2

2023-2024

Lahore University of Management Sciences

- Investigated cross-attention mechanisms in text-to-image diffusion models to improve generation quality and prompt adherence.
- Explored training-free guidance strategies for improving alignment between textual prompts and generated outputs.
- Studied how textual conditioning influences intermediate representations in diffusion-based generation pipelines.

Forecasting of Taxi Demand for an Urban Centre

2022-2023

Lahore University of Management Sciences

- Processed and visualized geo-temporal features for the taxi demand dataset provided by a local ride-sharing company (Careem) for a megacity (Lahore)
- Developed a joint architecture using LSTMs and Vision Transformers to model spatiotemporal demand patterns.
- Analyzed spatial and temporal factors affecting urban ride-hailing demand in a large metropolitan setting.

SELECTED COURSE PROJECTS

Robust Conformal Prediction with Geometric Intuition

Spring 2026

Rutgers University

- Studied theoretical foundations of conformal prediction and its coverage guarantees under distribution shift.
- Presented a geometric interpretation of conformal prediction sets to build intuition for uncertainty quantification in high-dimensional spaces.

Adam Optimizer: Visual and Mathematical Analysis

Fall 2025

Rutgers University

- Conducted a mathematical analysis of the Adam optimizer, including adaptive moment estimation, bias correction, and convergence behavior.
- Developed visual demonstrations to explain adaptive learning-rate behavior across loss landscapes.
- Studied counterexamples to Adam convergence and subsequent corrections proposed in later work.

Bilevel Optimization: Theory and Applications

Spring 2025

Rutgers University

- Investigated theoretical foundations of bilevel optimization, including optimality conditions, reformulations, and solution strategies.
- Studied applications of bilevel formulations in machine learning, optimization, and hierarchical decision-making.

Vehicle Detection and Classification from Image Data

Spring 2022

Lahore University of Management Sciences

- Processed more than 2,000 traffic-camera images for vehicle detection and classification.
- Implemented feature extraction techniques including HOG, edge detection, and Scale-Invariant Feature Transform.
- Programmed a sliding-window pipeline for vehicle localization using bounding boxes.

ADDITIONAL PROJECTS

Single-Chamber Microbial Fuel Cells in Urban Centres

[Water Informatics and Technology Center](#), LUMS

- Researched the effect of electrogens under varying substrates and experimental conditions.
- Built and deployed a working microbial fuel cell model with a live data-recording setup for field experimentation.

Traffic and Parking Data Collection for a Densely Populated Urban Region

Fall 2021

Lahore University of Management Sciences

- Collected traffic and parking data from a high-density urban region with more than 100,000 people per square kilometer.
- Processed and visualized collected data to support analysis for local urban policy planning.

TEACHING EXPERIENCE

Machine Learning for Engineers

Fall 2024, Fall 2025

Rutgers University | Instructor: Prof. Waheed U. Bajwa

- Delivered weekly recitations for more than 50 students, reinforcing key ideas and addressing conceptual gaps.
- Held office hours, prepared quizzes, graded exams, and reviewed semester-end course projects.
- Supported students with mathematical foundations and implementation details for machine learning models.

Programming Methodology II

Fall 2024

Rutgers University | Instructor: Prof. Yao Liu

- Conducted office hours, graded and proctored exams, and supported more than 100 students in a foundational programming course.
- Assisted students with programming concepts, debugging, and course assignments.

Engineering Laboratory

Spring 2023

Lahore University of Management Sciences | Instructor: Prof. Momin Uppal

- Conducted weekly laboratory sessions on fundamental electrical circuits for more than 100 students.
- Graded weekly lab manuals and evaluated comprehensive semester-end course projects.

Electromagnetic Fields and Waves

Fall 2022

Lahore University of Management Sciences | Instructor: Prof. Imran Cheema

- Designed, managed, and graded course components including weekly assignments and tutorials.
- Supported instruction for the undergraduate electrical engineering cohort.

Engineering Models

Fall 2021

Lahore University of Management Sciences | Instructor: Prof. Hasan Mohy-ud-Din

- Conducted weekly office hours and assisted students with core concepts in differential equations.
- Managed course components including quizzes, exams, and assignments.

HONORS AND AWARDS

Graduated with Distinction

2023

Lahore University of Management Sciences

Dean's Honor List of Academic Excellence

2019-2023

Lahore University of Management Sciences

LEADERSHIP AND SERVICE

President, ECE Graduate Student Association

2026-Present

Rutgers University

Treasurer, LUMS Anime Society

2022-2023

Lahore University of Management Sciences

Director of Events, LUMS Anime Society

2021-2022

Lahore University of Management Sciences

TECHNICAL SKILLS

Programming

Python, MATLAB, C++

Machine Learning and Data Science

NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow

Tools

LaTeX, Git, SolidWorks, Canva